ITA0312 -MOBILE COMPUTING

EXPERIMENT-12

12. Use Figma to create an infographic showing the evolution of mobile communication from 1G to 5G.

Aim:

To create an infographic using Figma showing the evolution of mobile communication from 1G to 5G, with the introduction period, major technology and key features of each generation.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame and give it the title “Evolution of Mobile Communication: 1G to 5G”.
3. Add five cards in sequence for 1G, 2G, 3G, 4G and 5G.
4. In each card, add the introduction period, major technology and key features of that generation.
5. Connect the cards using arrows to show the evolution from 1G to 5G.
6. Add a note below the cards explaining that the introduction periods are approximate.
7. Create a separate detail frame for each generation with its description, key features and engineering perspective.
8. Use a consistent colour scheme, icons and readable text across all frames.
9. Check all frames and save the Figma design.

Evolution of Mobile Communication: 1G to 5G

COMMUNICATION ENGINEERING

YEAR 2 / LAB ASSIGNMENT

Evolution of Mobile Communication: 1G to 5G

From analog voice calls to high-speed, low-latency connectivity.

1G

INTRODUCTION
Around 1980s
First commercial launch: 1979

MAJOR TECHNOLOGY
Analog cellular
AMPS and NMT

KEY FEATURES
Analog voice calls

→

2G

INTRODUCTION
Early 1990s
Introduced in 1991

MAJOR TECHNOLOGY
Digital cellular
GSM and IS-95 CDMA

KEY FEATURES
Digital voice and SMS

→

3G

INTRODUCTION
Early 2000s
Introduced in 2001

MAJOR TECHNOLOGY
UMTS / WCDMA
and CDMA2000

KEY FEATURES
Mobile internet and
video calling

→

4G

INTRODUCTION
Around 2010
Introduced in 2009–2010

MAJOR TECHNOLOGY
LTE / LTE-Advanced
using OFDMA

KEY FEATURES
High-speed internet
and HD streaming

→

5G

INTRODUCTION
Late 2010s
Introduced in 2019

MAJOR TECHNOLOGY
5G New Radio (NR)

KEY FEATURES
High-speed connectivity,
low latency, and
IoT support

Note: Introduction periods are approximate and vary by country. Sections are equally spaced and do not represent a time scale.

SMS = Short Message Service · OFDMA = Orthogonal Frequency Division Multiple Access · IoT = Internet of Things

1G — Generation detail

COMMUNICATION ENGINEERING / YEAR 2

1G · Detail

1G — Analog voice

APPROX. INTRODUCTION
Around 1980s
First commercial launch: 1979

MAJOR TECHNOLOGY
Analog cellular / AMPS and NMT

First-generation cellular networks made mobile voice calls possible using analog radio signals. Commercial cellular service began in Japan in 1979, followed by systems such as NMT and AMPS in the 1980s. Coverage was divided into cells so radio frequencies could be reused.

KEY FEATURES

- Analog voice calls over cellular radio networks.
- Limited capacity and security compared with later digital systems.

ENGINEERING PERSPECTIVE

1G established the cellular concept: frequency reuse and handover allowed calls to continue as users moved between cells.

2G — Generation detail

COMMUNICATION ENGINEERING / YEAR 2

2G · Detail

2G — Digital voice and SMS

APPROX. INTRODUCTION

Early 1990s
Introduced in 1991

MAJOR TECHNOLOGY

Digital cellular / GSM and IS-95 CDMA

Second-generation networks replaced analog voice transmission with digital encoding. GSM service launched in 1991; IS-95 CDMA followed later in the decade. Digital techniques improved spectrum use and enabled services such as SMS alongside voice calls.

KEY FEATURES

- Digital voice calls with improved capacity and security.
- SMS for short text messages between mobile users.

ENGINEERING PERSPECTIVE

GSM separates users using frequency and time slots; IS-95 uses code division. Both are digital cellular systems, but their radio-access methods differ.

3G — Generation detail

COMMUNICATION ENGINEERING / YEAR 2

3G · Detail

3G — Mobile internet

APPROX. INTRODUCTION

Early 2000s
Introduced around 2001

MAJOR TECHNOLOGY

UMTS / WCDMA and CDMA2000

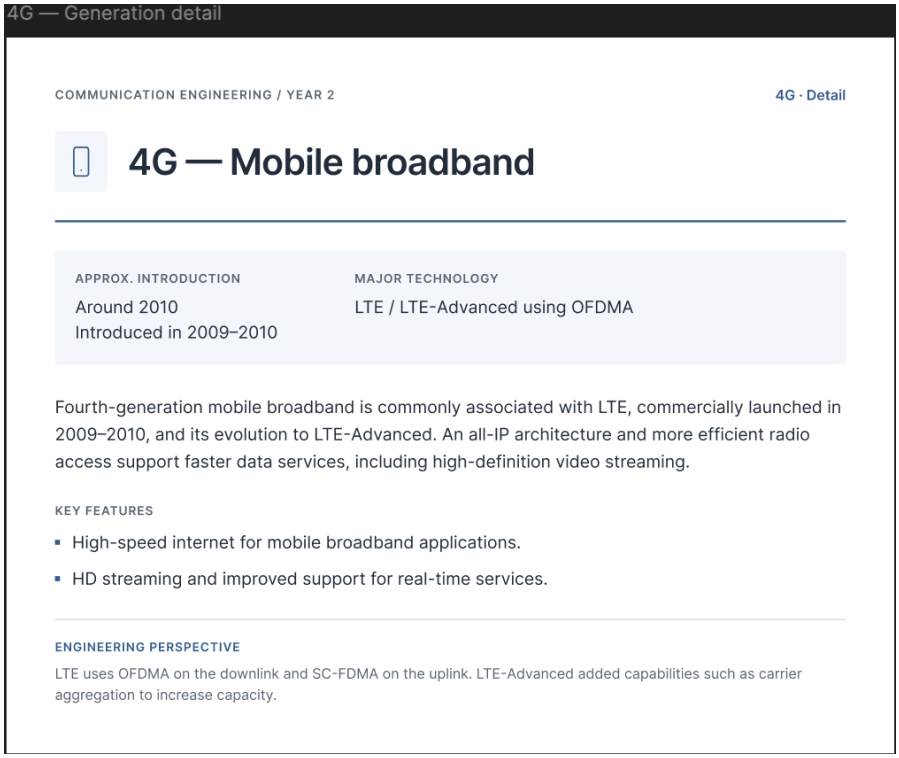
Third-generation networks expanded mobile communication beyond voice and text by supporting higher data rates. UMTS uses WCDMA for radio access, while CDMA2000 is another major 3G family. These networks made mobile web browsing and video calling more practical.

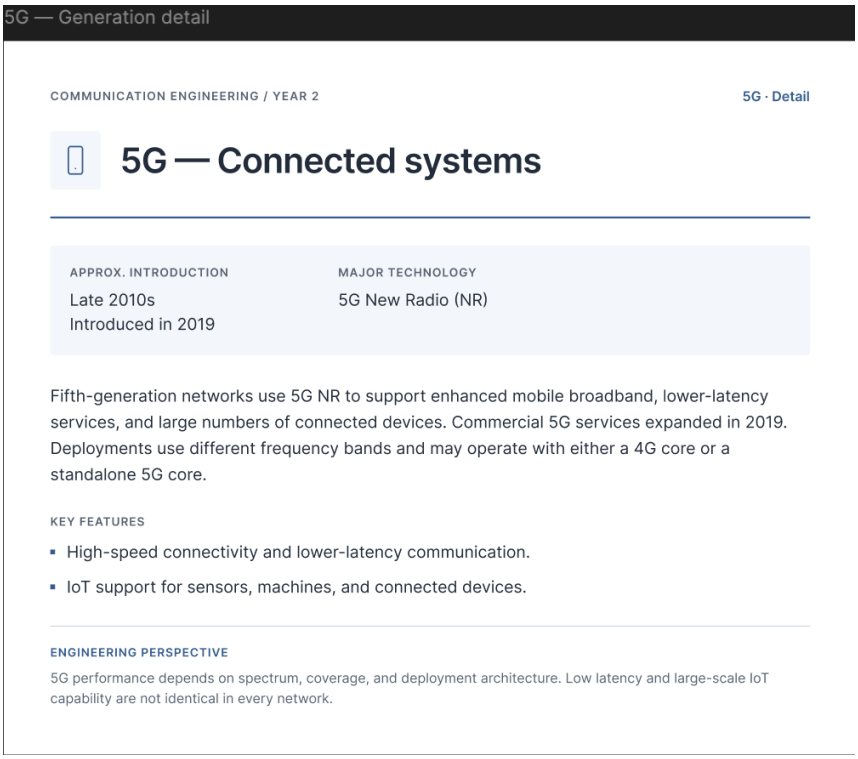
KEY FEATURES

- Mobile internet for browsing, email, and data services.
- Video calling and richer multimedia communication.

ENGINEERING PERSPECTIVE

Higher data capacity enabled new services, but actual throughput depended on the radio conditions, network load, and the specific 3G release.





Result:

An infographic showing the evolution of mobile communication from 1G to 5G, along with a detail page for each generation, was successfully designed using Figma.

EXPERIMENT-13

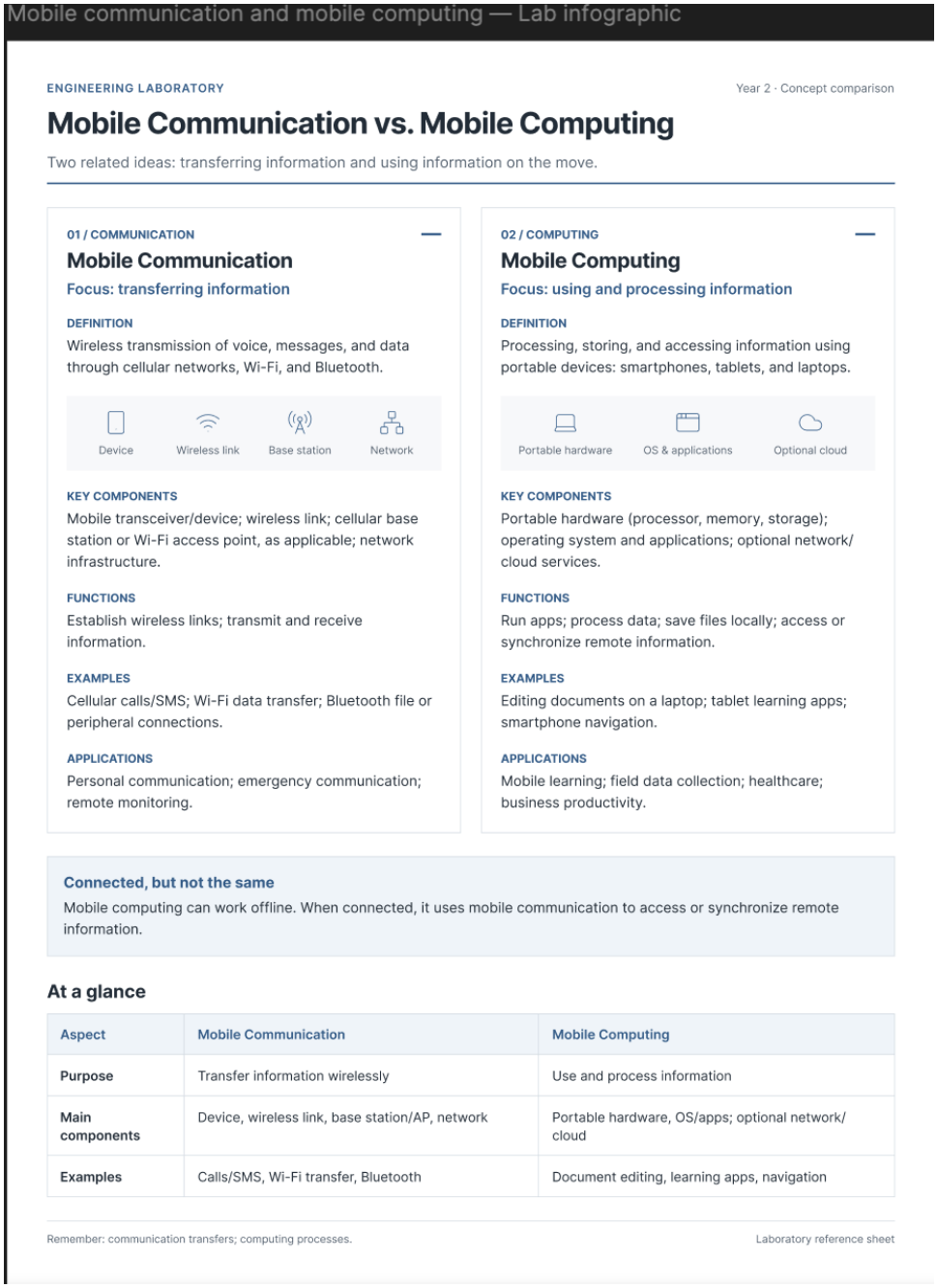
13. Use Figma to design an infographic comparing mobile communication and mobile computing.

Aim:

To design an infographic using Figma that compares mobile communication and mobile computing in terms of definition, components, functions, examples and applications.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame and give it the title “Mobile Communication vs. Mobile Computing”.
3. Add two cards side by side, one for Mobile Communication and one for Mobile Computing.
4. In each card, add the focus, definition, key components, functions, examples and applications.
5. Add icons representing the main components of each concept.
6. Add a highlighted note explaining how the two concepts are connected but not the same.
7. Add an “At a glance” comparison table covering purpose, main components and examples.
8. Use a consistent colour scheme and readable text, then save the Figma design.



Result:

An infographic comparing mobile communication and mobile computing was successfully designed using Figma.

EXPERIMENT-14

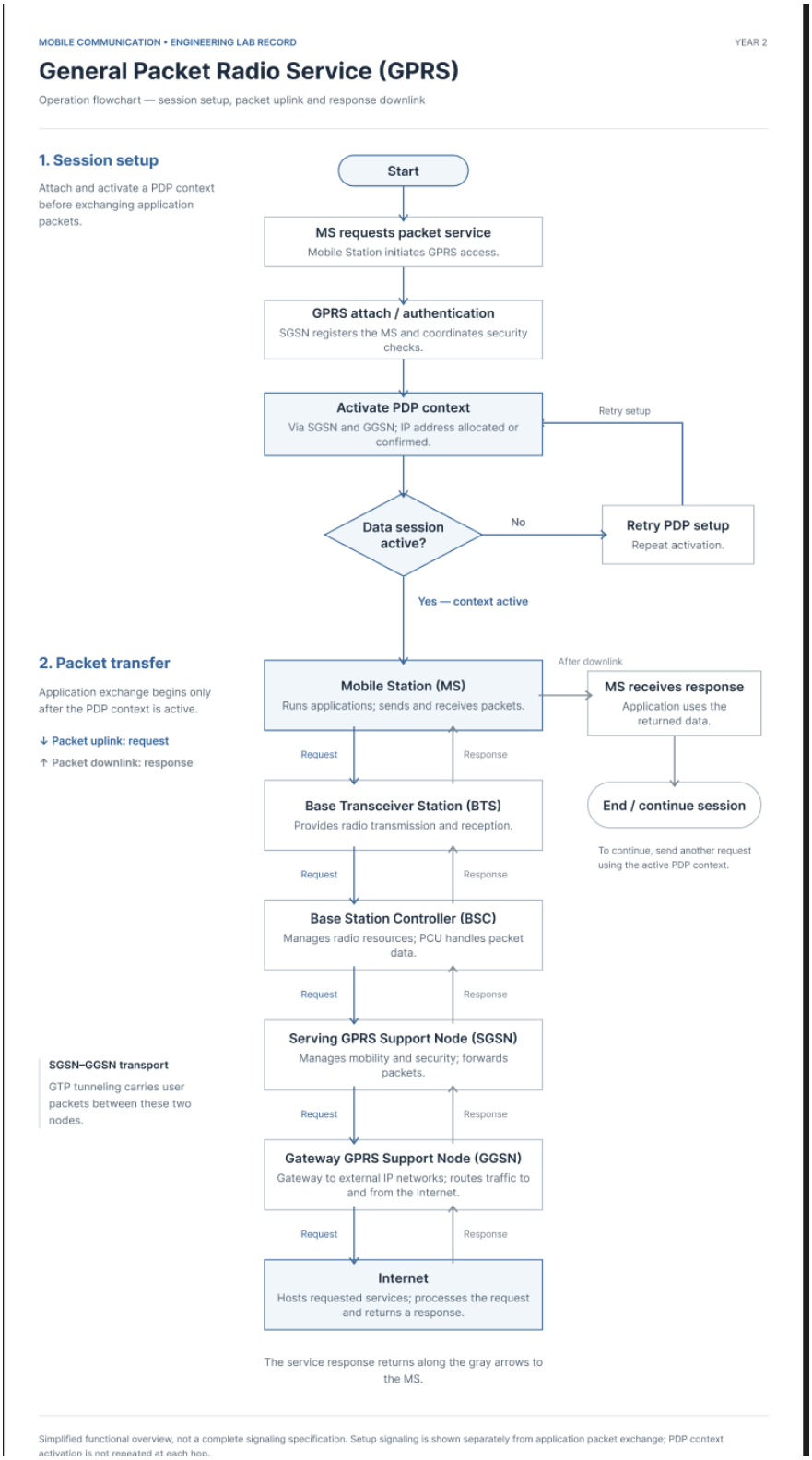
14. Develop a flowchart in Figma depicting the operation of the General Packet Radio Service (GPRS).

Aim:

To develop a flowchart using Figma depicting the operation of GPRS, including session setup, packet uplink and response downlink.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame and give it the title “General Packet Radio Service (GPRS)”.
3. Add flowchart shapes for the session setup: Start, MS requests packet service, GPRS attach / authentication, and Activate PDP context.
4. Add a decision diamond “Data session active?” with a retry path for PDP setup.
5. Add boxes for the packet transfer path: Mobile Station (MS), BTS, BSC, SGSN, GGSN and the Internet.
6. Connect the boxes using arrows to show the request (uplink) and the response (downlink).
7. Add the steps for the MS receiving the response and for ending or continuing the session.
8. Add labels and short descriptions, use colours to distinguish the components, and save the Figma design.



Result:

A flowchart illustrating the operation of GPRS, covering session setup, packet uplink and response downlink, was created using Figma.

EXPERIMENT-15

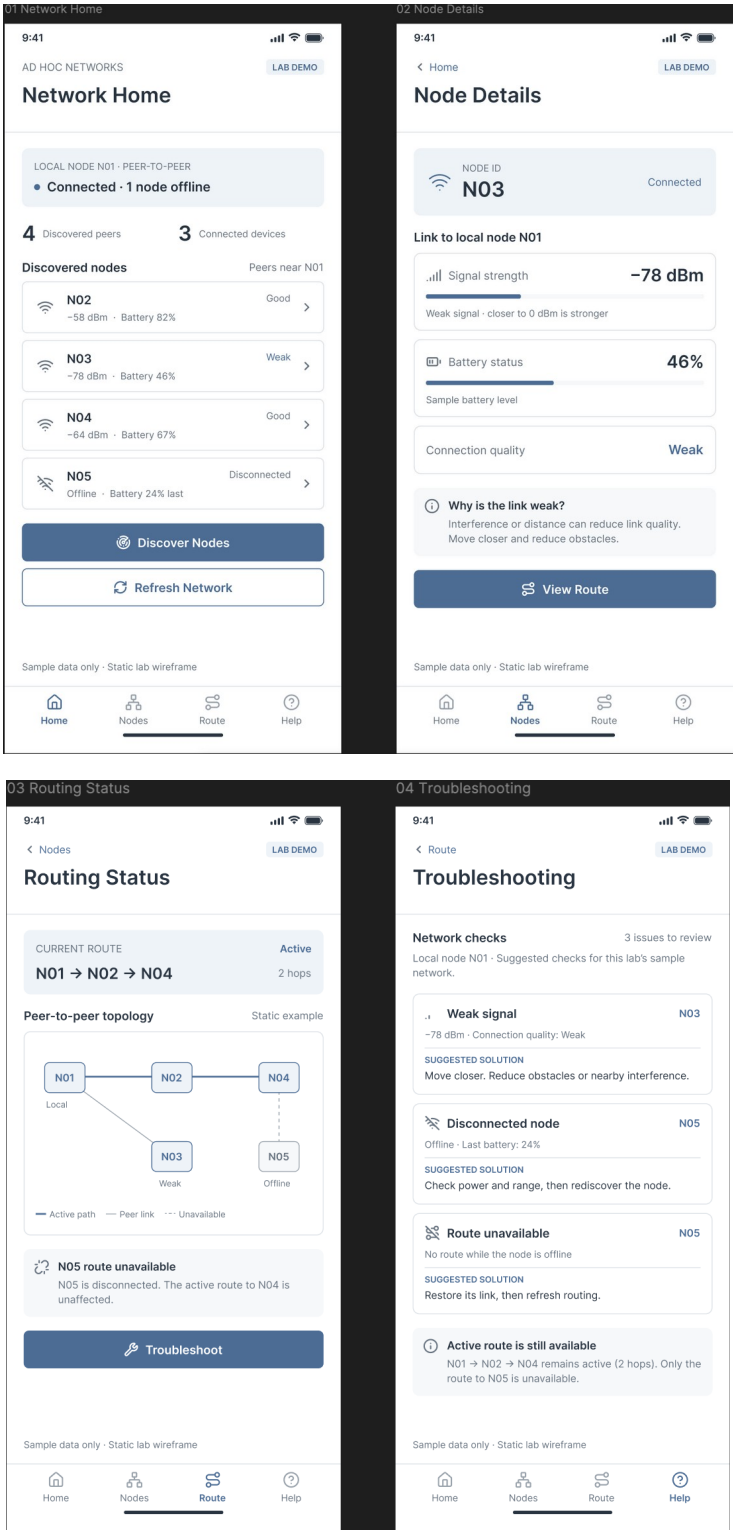
15. Design a mobile application prototype in Figma for monitoring an ad hoc network.

Aim:

To design a mobile application prototype using Figma for viewing the nodes, link quality and routes of an ad hoc network, and for troubleshooting network problems.

Procedure:

1. Open Figma and create a new design file.
2. Create mobile-sized frames representing smartphone screens.
3. Design the Network Home screen showing the local node, discovered peers, connected devices and the list of discovered nodes with signal strength and battery level.
4. Add the Discover Nodes and Refresh Network buttons.
5. Design the Node Details screen showing signal strength, battery status, connection quality and a View Route button.
6. Design the Routing Status screen showing the current route and the peer-to-peer topology.
7. Design the Troubleshooting screen listing network checks with suggested solutions.
8. Add a bottom navigation bar with Home, Nodes, Route and Help.
9. Connect the screens in Prototype mode and save the Figma design.



Result:

A mobile application prototype for monitoring an ad hoc network, with screens for network home, node details, routing status and troubleshooting, was successfully designed using Figma.